

Sean Lee

626-484-9787 | slee0339@usc.edu | Los Angeles CA

EDUCATION

Keck School of Medicine, University of Southern California

Los Angeles, CA

Doctor of Medicine

Expected Graduation May 2027

University of Southern California

Bachelor of Science, Human Biology. Healthcare Studies Minor

2019 - 2022

PUBLICATIONS

Cullaro G, Lai JC, Liu T, Gomez ME, Simonetto DA, Lee BP, **Lee S**, Allegretti AS, Mazumder NR, Patidar KR, Verna EC. Machine learning–derived clinical phenotypes of ascites in cirrhosis: A multi-center latent class analysis with external validation. *Journal of Hepatology*. 2026. (In revision for resubmission)

Elshafei M, Pouwels S, Putnam L, Munoz X, Eini JI, **Lee S**, Lipham JC, Houghton C. Implementation of a standardized protocol for EndoFLIP utilization alters hiatus closure intraoperatively: A proof-of-concept study. 2026. (In submission)

Meremikwu K, **Lee S**, Tang B, Wilson ML, Killu K. Diagnostic Accuracy of Lung Ultrasound for Pneumonia in the Intensive Care Unit: A Systematic Review and Meta-analysis. (In preparation)

RESEARCH

USC Keck School of Medicine

Los Angeles, CA

Department of General Surgery (GLS/MIS) | Dr. John Lipham, Dr. Luke Putnam

Research Assistant

Jan 2026 - Present

- Conducted retrospective chart review and data collection for a study evaluating the impact of intraoperative platelet-rich plasma (PRP) injection on postoperative outcomes following hiatal hernia repair.
- Performed retrospective chart review and data collection evaluating intraoperative EndoFLIP utilization during antireflux surgery and its impact on real-time surgical decision-making and adjustment.
- Prepared results section and contributed to full manuscript development examining whether EndoFLIP-guided intraoperative measurements reduce postoperative reflux and dysphagia outcomes.

Department of Pulmonary & Critical Care | Dr. Keith Killu

Nov 2025 - Present

Project Lead | Research Assistant

- Conceived and designed an independent retrospective study examining clinical and laboratory predictors of imaging escalation from chest X-ray to CT in hospitalized pneumonia patients, with the goal of developing evidence-based criteria to guide resource-appropriate workup.
- Formulated study hypothesis, defined patient cohort inclusion/exclusion criteria, selected candidate clinical variables, and prepared IRB submission materials.
- Lead all phases of the project including data collection, statistical analysis, and manuscript preparation for peer-reviewed publication.
- Conducted comprehensive systematic literature review and study screening for a meta-analysis comparing diagnostic accuracy of point-of-care ultrasound versus CT for pneumonia, identifying and evaluating studies across multiple databases.
- Extracted and synthesized quantitative data across included studies, assessing methodological quality and pooling effect estimates for analysis.
- Contributing to manuscript drafting and preparation for peer-reviewed submission.

Department of Transplant Hepatology | Dr. Brian Lee, Dr. Giuseppe Cullaro

Research Assistant

Sept 2024 - Present

- Performed detailed chart review and longitudinal data extraction for 150+ liver transplant patients spanning pre-listing through post-transplant outcomes, requiring individualized manual review of complex transplant records.
- Contributed to a study evaluating Cystatin C-based MELD scoring as a refinement to standard prognostic indices for improving post-transplant outcome prediction.
- Performed chart review and data extraction for a multi-center study identifying clinical predictors of cirrhosis and ascites phenotypes, contributing to a dataset spanning multiple institutions.
- Contributed to manuscript development and review examining phenotype-based risk stratification in cirrhosis; manuscript currently under revision for peer-reviewed publication.

Department of Emergency Medicine | Dr. Alexander Sapin

Research Assistant

April 2026 - Present

- Contributed to a quality improvement initiative aimed at developing an automated flagging system to identify ED patients at risk for prolonged length of stay (LOS >21 days) to facilitate early disposition planning.
- Performed data collection and analysis of admission diagnostic codes for patients meeting LOS threshold, forming the evidential basis for future flag criteria development.

USC Alfred E. Mann School of Pharmacy | Dr. Yong (Tiger) Zhang*Research Assistant***Los Angeles, CA**

June 2020 - June 2022

- Conducted weekly literature reviews screening 200+ papers to identify novel antibody candidates for smart-exosome-based anticancer drug delivery research.
- Performed protein and DNA purification, cell culture maintenance, and preparation of Antibody-Drug Conjugate specimens in support of CD36-targeted cancer antibody studies.

PERTINENT EMPLOYMENT HISTORY**AHMC: Garfield Medical Center****Monterey Park, CA***Patient Liaison | Project Manager*

May 2022 - May 2023

- Collaborated with attending physicians and nursing staff to triage and resolve urgent patient needs in real time, contributing to measurable improvements in patient satisfaction and care continuity.
- Served as a real-time medical interpreter across Cantonese, Mandarin, and English, facilitating accurate communication between patients, families, and clinical teams during high-stakes encounters including diagnoses, discharge planning, and informed consent.
- Performed targeted literature reviews synthesizing emerging COVID-19 evidence to support bi-weekly grand rounds presentations attended by medical and administrative staff across the health system.
- Aggregated and analyzed HCAHPS, Leapfrog, and ED-Throughput metrics across a 9-hospital system, translating performance data into actionable summaries to guide senior leadership's operational decisions.
- Drafted and distributed formal meeting minutes for senior administrative and clinical leadership, ensuring accurate documentation of decisions, action items, and follow-ups across high-level institutional meetings.

LANGUAGES

English – Native

Chinese Cantonese – Conversational proficiency

Chinese Mandarin – Intermediate (able to take patient history and perform physical exam)

Spanish – Intermediate (able to take patient history and perform physical exam)

LEADERSHIP/VOLUNTEER EXPERIENCE**USC Keck School of Medicine****Los Angeles, CA***Office of Student Affairs | Student Affairs Representative*

Aug 2023 - Aug 2025

- Served as a primary liaison between the student body and institutional administration, advocating for student needs and driving improvements in campus resources and support services.
- Partnered with administrators to identify, propose, and implement targeted solutions addressing student well-being, including housing equity, facility enhancements, and mental health resource accessibility.
- Cultivated an inclusive environment that encouraged student engagement and empowered peers to contribute meaningfully to institutional decision-making and curricular development.

Gastroenterology Student Interest Group | Vice President

Aug 2023 - Dec 2024

- Spearheaded programming initiatives by coordinating with faculty and clinical speakers to curate lunch talk series, delivering career-relevant insights to students pursuing medicine.
- Led targeted outreach to identify and compile a diverse roster of research-active faculty, connecting students with mentorship and research opportunities across a broad range of subspecialties and interests.

USC Photography Club | President | Events Director

Aug 2020 – Aug 2023

- Led a 15-person executive team to organize events, retreats, and workshops for 2,000+ members, overseeing event planning, marketing, and logistics operations.

Dmanisi Archaeological Field School**Dmanisi, Georgia***Field Researcher*

Jun 2019 - Aug 2019

- Participated in active excavation at the Dmanisi site, one of the most significant *Homo erectus* fossil sites outside Africa, gaining hands-on exposure to paleoanthropological methods and human evolutionary history.
- Attended lectures on hominin morphology, species differentiation, and the anthropological significance of early human migration.

Alhambra Hospital**Alhambra, CA***Emergency Medicine Shadowing | Volunteer*

July 2020 – Dec 2022

- Observed clinical decision-making in high-acuity emergency settings across diverse patient populations, and assisted nursing staff with patient care, vitals, and unit operations.

Connect Tech Network**San Francisco, CA***Tutor Volunteer*

June 2020 – June 2021

- Taught digital literacy (Zoom, Google tools) in Cantonese to elderly clients during COVID-19 to improve accessibility of services.

PROJECTS

AI-Powered Clinical Documentation Tool (Independent Project)

April 2026

- Built a HIPAA-compliant web tool using keyword recognition and speech dictation to generate structured clinical notes, standardizing documentation output across encounter types without retaining patient data.

[\[https://github.com/infer321/note-helper/\]](https://github.com/infer321/note-helper/)

Antibiotic Stewardship Tool (Independent Project)

May 2026

- Developed a publicly deployed web-based prescribing reference covering 30+ infection types across 9 organ systems, with dynamic adjustments for penicillin allergy and MRSA risk; delivers setting-appropriate regimens (outpatient/floor/ICU) aligned with current IDSA/ASHP guidelines in 3 clicks.

[\[https://github.com/infer321/AntibioticStewardshipTool/\]](https://github.com/infer321/AntibioticStewardshipTool/)

Internal Medicine Clinical Scoring Calculator (Independent Project)

May 2026

- Built a mobile-optimized, offline-capable web calculator consolidating 15 validated scoring tools (HEART, CURB-65, MELD-Na, Wells PE/DVT, SOFA, APACHE II, and others) with real-time interpretation and evidence-based clinical guidance.

[\[https://github.com/infer321/InternalMedCalc/\]](https://github.com/infer321/InternalMedCalc/)

Rounds (Independent Project)

May 2026

- Designed and developed a cross-platform mobile app (iOS/Android) in React Native to reduce cognitive load during clinical rotations, featuring patient task tracking, clinical timers, and shift signout checklists; no patient data stored.

[\[Rounds, App Store Download\]](#)

CERTIFICATES

Quality Improvement Certificate | USC Keck School of Medicine

2026

- Completed formal training in QI methodology, including process improvement frameworks and patient safety principles applicable to clinical practice.

Certificate of Merit Piano, Level 10 (Highest Level) | Music Teachers' Association of California

2018

- Achieved the highest level of the statewide classical piano assessment, demonstrating years of disciplined technical and musical development

PROFESSIONAL INTERESTS

Piano, Weightlifting, Cycling, Running, Photography.